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Title

Great Lakes Ice Phenology: A New Model to Examine Changing Cover and Variability

Authors

Sabrina Jaffe[1], Lacey Mason[2], Craig A. Stow[2], Song S. Qian[3]

Affiliations

[1]Department of Biological and Environmental Sciences, Wittenberg University, Springfield, OH

[2]Great Lakes Environmental Research Laboratory, National Oceanic and Atmospheric Administration, Ann Arbor, MI

[3]Department of Environmental Sciences, The University of Toledo, Toledo, OH

Corresponding Author

Sabrina Jaffe

Author Contributions

Sabrina Jaffe - Study design, performed the analysis, wrote the draft and final version of the manuscript
Lacey Mason - Study design, contributed data, manuscript preparation
Craig A. Stow - Study design, contributed data, manuscript preparation
Song S. Qian - Conceptualization, study and data analysis design, manuscript preparation

Key Points

1. A smoothed hockey stick model within the Bayesian hierarchical model framework summarizes annual Great Lakes' ice cover patterns.
2. Changes in peak ice cover, first and last day of 10% ice cover, duration, and variability were found using the model.
3. This model provides a consistent approach for characterizing a wide range of annual responses that follow a unimodal pattern.

Key Words

Great Lakes, Ice Cover, Bayesian Statistics, Hierarchical Model

Abstract

A modeling framework that characterizes annual patterns can be advantageous for evaluating different features of the data over time. For instance, a modeling framework for evaluating changes in seasonal patterns of physical and biological features over time can be a useful tool for studying the effects of climate change. We introduce a smoothed hockey-stick model and a Bayesian hierarchical modeling approach to summarize yearly Laurentian Great Lakes ice cover patterns, allowing detailed characterization of them and their trends. We found that the first and last days of at least 10% ice cover have shifted later and earlier, respectively, resulting in decreased duration. Furthermore, ice cover variations (day-to-day variation within a year and peak coverage variation over time) have increased and peak ice coverage have decreased over the study period. The effects of a reported regime shift that occurred after the 1997 – 1998 winter are largely reflected in the peak-ice cover, with reduced peak ice coverage and a higher likelihood of little to no significant ice cover after the shift. We demonstrated the effectiveness of the Bayesian hierarchical modeling approach in summarizing the variation in ice cover patterns, providing insight for future climate impact studies and management strategies. This modeling approach can be applied to a wide range of phenomena that follow a similar annual pattern.

Plain Language Summary

This paper presents a flexible model for summarizing yearly patterns for many natural phenomena. Using lake ice cover data of the Great Lakes, we demonstrate the utility of such a model. Lake ice cover influences lake ecosystems, and many communities rely on ice cover for social, economic, and recreational purposes. Given the ecological and

societal impacts of lake ice, it is important to understand how ice cover has changed in the past and what to expect for the future. Our analysis provides a complete description of ice cover annual patterns of the five Great Lakes, including when ice appears, peaks, and disappears, and how the timings of these important events have changed over the years. We found that overall, we have less ice cover for a shorter period each year. In addition, the variation of ice cover has increased over time. Our approach provides a tool for evaluating changes in ice cover behavior, helping improve future monitoring of Great Lakes ice cover. Our model can be used for a wide variety of cases where the yearly pattern behaves similarly to ice cover.

1. Introduction

Lake physical, biological, and chemical responses can serve as important climate sentinels (Adrian et al., 2009). Climate shifts impact freshwater lakes, including the Laurentian Great Lakes (Figure 1; Karl et al., 2009; Woolway et al., 2020; Abdelhady et al., 2025), hereafter the Great Lakes. One important condition in the Great Lakes that has been changing is ice cover. Ice cover and duration is decreasing in northern latitude lakes (Woolway et al., 2020; Basu et al., 2024; Walsh et al., 2025) and as global temperatures increase there will be more lakes in the northern hemisphere that experience intermittent ice cover (Sharma et al., 2019). In the Great Lakes region, ice cover duration and annual maximum have decreased (Assel et al., 2003; Jensen et al., 2007; Wang et al., 2012; Mason et al., 2016;). Additionally, freezing dates have become later and ice breakup dates earlier in northern hemisphere lakes including the Great Lakes region (Magnuson et al., 2000; Jensen et al., 2007; Sharma et al., 2021). In addition to these trends in ice cover, a regime shift in the Great Lakes region was documented following the warm El Niño winter of 1997 – 1998, marking a point of noticeable changes in ice cover patterns including decreased ice cover and increased annual variation (Van Cleave et al., 2014, Lin et al., 2022; Venumuddula et al., 2024). A lake's response to changing temperature will differ based on factors such as latitude and longitude, elevation, and morphology (Woolway et al., 2020; Sharma et al., 2021). Responses can also differ within a lake. For instance, nearshore and offshore regions may experience different rates of change (Mason et al., 2016). Ultimately, changing ice phenology will have environmental and societal consequences (Hampton et al., 2024; Hernández-Carrasco et al., 2025).

Ice cover is a key variable that controls ecosystem function (Ozersky et al., 2021), and its phenological changes could impact under ice ecology. For instance, in northern hemisphere lakes, winter stratification is projected to decrease (Woolway et al., 2022) which could impact nutrient and oxygen circulation patterns. Wind in areas of low ice coverage can disrupt thermal stratification by mixing the water column, causing phytoplankton to become vertically redistributed (Beall et al., 2016). The presence and thickness of ice and snow cover will influence available light, impacting winter primary productivity rate (Wetzel, 2001). Additionally, decreasing ice cover can negatively impact native fish species, such as the lake whitefish (*Coregonus clupeaformis)* that rely on the ice to protect eggs from wave damage (Lynch et al., 2010). Because ice cover acts as a barrier between the lake and atmosphere, reduced ice cover could enhance heat and moisture exchange in the winter (Gerbush et al., 2008). However, some studies have also suggested that the overall effect of ice cover on evaporation is relatively small (Anderson & Gronewold, 2025). Earlier ice-off can also expose the water to solar radiation for a longer period affecting lake warming rates (Li et al., 2022). Changes in ice cover patterns will also impact society as many communities rely on lake ice for economic, social, and cultural purposes (Knoll et al., 2019; Filazzola et al., 2024, Hampton et al., 2024). For instance, ice cover affects commercial shipping (Rust and Steward, 2024) and recreational activities such as ice fishing and skating. With reduced ice cover, activities that depend on ice, like skating, will become more limited, whereas shipping may not be as restricted in the winter. Reduced ice cover can also increase safety risks by exposing larger areas of open water to cold atmospheric conditions which can enhance lake-effect snowfall (Burnett et al., 2003, Hampton et al., 2024).

Given the major ecological and societal implications of ice cover, it is important to determine how ice phenology is changing due to global temperature changes.

Previous studies have investigated changes in lake ice cover based on annual statistics calculated from daily observations (Wang et al., 2012; Van Cleave et al., 2014; Mason et al., 2016; Sharma et al., 2021). For example, Wang et al. (2012) characterized the seasonal cycle of Great Lakes ice cover using weekly ice area and its interannual variability (expressed as the standard deviation across years for each week). They also estimated annual mean ice cover by averaging ice data over each ice year and used least-squares regression to assess long-term trends in annual ice cover. Van Cleave et al. (2014) calculated 15-day running means of ice cover for Lake Superior to derive several ice metrics, including ice-on and ice-off dates, duration, and maximum ice cover. They then analyzed temporal changes using linear regression and step-change analyses to assess the full time period and decadal-scale variability, respectively. These studies have provided important insights into changing ice phenology, but such approaches typically require calculating each metric separately and do not fully characterize the underlying annual pattern of ice cover. When evaluating long-term trends, the primary focus should be on long-term patterns rather than daily fluctuations. Non-parametric methods often lack readily interpretable parameters and may require fitting separate models for individual years, limiting consistency across time. In contrast, parametric approaches provide a consistent modeling framework with interpretable parameters, making them well suited for assessing long-term trends in phenological phenomena (Maguire et al., 2024).

For our study, we introduce a parametric model that summarizes the complete annual cycle of Great Lakes ice cover. A key advantage of this approach is that a single flexible model can generate a wide range of phenological metrics, and these metrics can be compared across years to evaluate temporal changes in ice phenology. By using a mathematical function with a limited number of parameters to describe the overall shape and timing of each year's ice cover pattern, we can derive many useful summary statistics, which provide a more complete description of annual ice dynamics and their temporal changes. To demonstrate the versatility of this approach, we derive several characteristics of Great Lakes annual ice cover that can be obtained from the model including peak ice cover, day of peak ice cover, and timing. This approach differs from previous methods because all of the ice cover metrics can be derived for each ice year in a consistent way using the estimated parameters from a single model. We also adapt the model to evaluate the changes in ice cover that occurred after the documented regime shift following the strong El Niño winter of 1997 – 1998, allowing us to quantify the changes in ice phenology following this shift. The model is implemented within a Bayesian hierarchical modeling (BHM) framework, which allows information to be pooled across years and reduces parameter estimation uncertainty (Gelman et al., 2014). The BHM approach has been applied in studies related to the cryosphere (Gopolan et al., 2018). By incorporating a flexible parametric model under a BHM framework, we can better quantify the nature of phenological changes in Great Lakes ice cover, which can improve our understanding of a variety of lake processes (Wilkinson et al., 2026). While we use the model to evaluate Great Lakes ice cover, this

approach has the potential to capture the annual pattern of a wide range of phenomena that follow a similar unimodal yearly cycle.

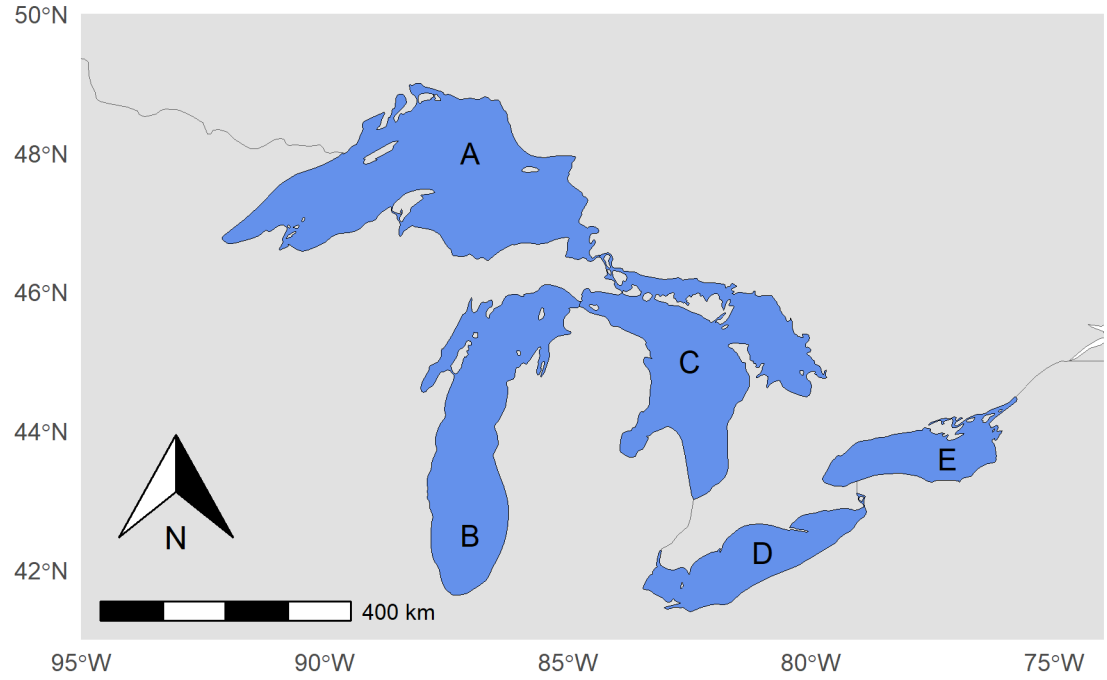


Figure 1. Map of the Laurentian Great Lakes: (A) Lake Superior, (B) Lake Michigan, (C) Lake Huron, (D) Lake Erie, and (E) Lake Ontario.

2. Methods

2.1 Laurentian Great Lakes Ice Cover Data

We obtained daily average lake-wide percent ice cover data from the National Oceanic and Atmospheric Administration's (NOAA) Great Lakes Environmental Research Laboratory (GLERL) for lakes Superior, Michigan, Huron, Erie, and Ontario (Figure 1, NOAA, 2020; NOAA, 2025). A detailed explanation of the data history and

processing methods are found in Yang et al. (2020). All five datasets span 1973 – 2024, and are organized by ice year, where the first day is November 10th and the last day is June 5th, except for Lake Erie where the last day is June 4th. Yearly patterns varied across lakes and years.

2.2 Smoothed Hockey Stick Model

Great Lakes ice cover is a seasonal phenomenon with a cyclical, annual pattern, moving from its annual low to its annual high and back again. This unimodal pattern can be described using a general function across time consisting of a rising arm that smoothly transitions into a declining arm. One such function is the smoothed hockey-stick model, a generalization of the piece-wise linear model often used in statistical change point analysis (Stephens, 1994; Qian and Richardson, 1997). The piece-wise linear model consists of two lines with different slopes, meeting at the change point.

$$y_{response} = \{ \quad a + \varepsilon, \quad x < \phi \; a + \beta\left(x_{time} - \phi\right) + \varepsilon, \; x \geq \phi \qquad (1)$$

where y is the response, α is the y-intercept of the left line segment, β is the slope of the right line segment, x is the time variable, ϕ is the changepoint, and ε is the model error term. Because the piecewise linear model from Equation 1 has an abrupt slope change (discontinuous first derivative), numerical difficulties arise when the model fits to noisy data (Chiu et al., 2006). Qian et al. (2022) used the sigmoid four-parameter logistic function to smooth the abrupt change in the first derivative and derived the smoothed version of the piecewise linear model (Equation 2):

$$y = \beta_0 + \beta_1\left(x - x_{mid}\right) + \delta\, scale^{\left(\frac{x - x_{mid}}{scal}\right)} + \varepsilon \qquad (2)$$

In our application, *y* is the logit transformed percent ice cover at day *x*, x_{mid} is the day of peak ice cover and ε is the error term associated with the model. Because the logit transformation is undefined for values of 0 and 1, observations with 0 ice cover were excluded, and observations with 1 were adjusted to 0.975 using the 'logit()' function in R prior to model fitting. Equation (2) alters the initial piece-wise linear model (Equation 1). Under this equation, $\delta = \beta_2 - \beta_1$, where β_1 and β_2 are related to the rates of increasing and decreasing ice cover, respectively, and β_0 is the peak ice cover at x= x_{mid}. *scal* is a smoothness parameter that determines the degree of bending of the two linear segments in the piecewise linear model. Because *scal* and δ are numerically non-identifiable (Qian et al., 2023), the value of *scal* is often fixed as the range of the smooth transition from one slope to the other. The larger the smoothness parameter (*scal*), the flatter the fitted curve will be. For *scal*, we used a fixed value of 0.1 which allowed for the transition region to span approximately 10% of the duration between the first and last day of ice cover in each year. As long as we are consistent with our selection of *scal*, then the model is reproduceable.

For ice cover, the annual pattern is not necessarily symmetrical for a given year. The smoothed hockey stick model is flexible in its ability to capture the asymmetry of the unimodal pattern from year to year. When applied to an appropriately transformed response variable, we can use this fitted parametric function to derive quantities that characterize the underlying patterns, either using analytical formulas (e.g., the logit of

the peak ice cover is $\beta_0 + \delta\, scallog(2)$) or numerical search methods (e.g., the first and last days of at least 10% ice cover, from which we estimated the ice cover duration). With these estimated characteristics, we can present the temporal trends for each lake.

2.3 Bayesian Hierarchical Model

Because we have data for individual years, the ice cover datasets are well suited for applying a Bayesian hierarchical modeling (BHM) framework within each lake, which allows us to leverage parameter information across years. We specify the annual model (Equation 2) as year-specific with model coefficients indexed by year (j):

$$y_j \sim Normal(\mu_j,\ \sigma_j)$$

$$\mu_j = \beta_{0j} + \beta_{1j}(x_j - x_{midj}) + \delta_j scal_j e^{(\frac{x_j - x_{midj}}{scal_j})} \qquad (3)$$

Equation (3) was estimated for each lake independently. We imposed a common prior distribution on the model coefficients to reflect the hierarchical structure of the data. Coefficients β_0, β_1, and β_2 are regression coefficients and a normal prior is reasonable. The change point x_{mid} is the peak day, which is unlikely to be a normal random variable. We transform x_{mid} by using the fraction of days prior to the peak day as a fraction of the ice season duration (γ_j), and impose a normal prior on the logit transformed γ_j:

$$logit(\gamma_j) = log(\frac{\gamma}{1-\gamma_j})$$

$$\beta_{0j},\ \beta_{1j},\beta_{2j},\ logit(\gamma_j) \sim Normal(\mu_\theta,\ \sigma_\theta) \qquad (4)$$

Where μ_θ is the mean for coefficients over many years and σ_θ^2 is the among year variance. The common prior links ice data from all 52 years which induces the shrinkage effect. For each year, x_{mid} was calculated by:

$$x_{midj} = D_{firstj} + \left(D_{lastj} - D_{firstj}\right) * \gamma_j \quad (5)$$

Where D_{first} and D_{last} represent the first and last day of observed ice for a given year. The models were fit using Markov chain Monte Carlo simulation using Stan (Stan Development Team, 2024) in R (R Core Team, 2023).

2.4 Evaluating changes in yearly patterns

Using the estimated parameters from the fitted curves for each year, we derived metrics representing important features of the ice cover seasonal patterns, including the day of the curve peak (x_{mid}) and the ice cover at the curve peak for each year. These metrics were selected because changes in the timing of the peak can indicate shifts in when maximum ice cover occurs, while changes in peak ice cover can reveal long-term changes in the magnitude of maximum ice extent. We also estimated the first and last days when ice cover reached 10% and calculated the duration (in days) of at least 10% ice coverage for each year. This allowed us to evaluate changes in both the timing and persistence of seasonal ice cover. Some years were unable to estimate the duration due to the estimated annual ice pattern not reaching 10% except for 1990 when the cover pattern was bimodal for Lakes Erie and Michigan (Figure S1). Because the first day of 10% ice cover could not be estimated for this year for Lakes Erie and Michigan,

we excluded the year 1990 for the 10% ice cover analysis for Lake Erie and Lake Michigan. For the years where the model fit well but we could not estimate 10% ice cover (indicating that 10% ice cover was not reached for a year), we used fixed values of 150, 50, and 0 for first day, last day, and duration, respectively. These values for first and last day were chosen to allow these years to be included in the evaluation while clearly indicating that the 10% ice cover threshold was not reached, which is reflected by the duration of 0.

Sen's slope was used to calculate the linear rate of change because this nonparametric method uses the median of all pairwise slopes to determine the rate of change and is less influenced by extreme values (Sen, 1968). We calculated Sen's slope for peak cover, peak day, first day, last day, and duration to determine the rate of change and the strength of the trend for each lake and each metric. This analysis informs us how these parameters have changed over time.

Ice cover variability is estimated using the differences (residuals) between the observed ice cover data and the respective model estimations. We estimated annual standard deviations (σ_j in Equation 3) for each lake.

2.5 Evaluating the Regime Shift

Our model can be used to provide specific changes due to the documented regime shift that occurred following the warm El Niño winter of 1997 – 1998 (Van Cleave et al., 2014, Lin et al., 2022; Venumuddula et al., 2024). To model the shift, we added a

parameter to the model to allow model coefficients to differ before and after the regime shift:

$$\beta_{0j},\ \beta_{1j},\beta_{2j},\ logit(\gamma_j) \sim N\left(\mu_\theta + R * a_\theta,\ \sigma_\theta\right) \qquad (5)$$

Where $R = 0$ and $R = 1$ for indicating before and after regime shift, respectively, and a_θ is the estimated average difference for each parameter before and after regime shift for each coefficient. The greater the difference in a's, the greater the pre- and post-regime shift change for each parameter. We applied this method to three different regime shift breakpoints: ice years 1997, 1998 (the breakpoint documented by Van Cleave et al. (2014) and Lin et al. (2022)), and 1999. This allowed us to compare the average differences of three breakpoints to determine which ice year better splits the data. We reported the results of the 1998 breakpoint in the Results section because this was the ice year reported by Van Cleave et al. (2014) and Lin et al. (2022). Results for the 1997 and 1999 breakpoints can be found in Table S1. This analysis allowed us to see how the regime shift is reflected in the parameters of our model to better understand the nature of the observed shift.

3. Results

Each year has its own fitted curve for all five lakes (Figure S2). There were no discernable changes in the study period in peak day for any of the five lakes (Table 1, Figure S3), which suggests that the timing of when peak ice cover occurs has not changed. For peak cover, Lakes Superior, Michigan, Huron, and Ontario all had significant decreases (Sen's Slope different from zero) while Lake Erie was decreasing

but did not have a significant trend (Sen's Slope not different from zero; Table 1). However, Lake Erie's peak ice cover exhibits a notable increase in annual fluctuation after around 1998 (Figure S4). All the lakes except Lake Erie have significant positive trends in the first day of at least 10% ice cover (Figure S5), which suggests that this first day is occurring later in the year. All five lakes have significant negative trends for the last day of 10% ice cover (Figure S6) which suggests that the timing of this last day occurs earlier in the season. Consequently, all five lakes experienced decreased duration of 10% ice cover and increased frequency of years with peak cover less than 10% since 1998 (Table 2, Figures 2, S5, and S6), indicating a decline in the number of days with ice cover of at least 10%. The decrease in duration was greatest for Lake Superior, which decreases on average by 1.4 days/year (Table 2).

We calculated daily residuals using the estimated daily mean coverage and observed coverage (Figure 3). Lake Superior maintains a stable variance across the 52 years while Lakes Michigan and Huron have steadily increasing variance. Lakes Erie and Ontario have elevated variance in more recent years. Increasing variance indicates that ice cover has become more unpredictable from day to day.

For evaluating the regime shift, the biggest change is reflected in coefficient β_0 reflecting the peak cover. The peak cover is approximately stable before and after 1998, but all lakes have a decreased average ice cover, suggesting a step change pattern (Figure 4). There was no difference before and after 1998 for other coefficients representing peak day (x_{mid}), rate of increasing (β_1), or the rate of decreasing (β_2, Table S1). Using the model, we determined that the shift occurred shortly (1 – 2 years) before

2000 with the main difference being the decreased peak ice coverage and increased likelihood of no significant ice cover (10%) after the shift. Additionally, most years where duration, first, and last day could not be estimated occur after the shift (Figures 2 and S6).

Table 1. Trend and P-value from the Sen's slope analysis for peak day (φ; days/year) and ice cover at curve peak (% cover/year) for all five Great Lakes. Significant trends (P-value < 0.05) are indicated with an asterisk.

	Peak Day		Peak Cover	
Lake	**Trend**	**P-value**	**Trend**	**P-value**
Erie	0.04	0.89	-0.37	0.12
Superior	-0.01	0.91	-0.77*	4.73e-4
Michigan	0.08	0.71	-0.35*	0.02
Huron	0.04	0.79	-0.56*	0.02
Ontario	0.38	0.07	-0.25*	4.85e-3

Table 2. Trend, indicating days/year, and P-value from the Sen's slope analysis for duration of at least 10% ice cover and first and last day of 10% ice cover for all five Great Lakes. Significant trends (P-value < 0.05) are indicated with an asterisk. The year 1990 was excluded from the Lake Erie and Lake Michigan analysis.

	Duration		First Day		Last Day	
Lake	**Trend**	**P-Value**	**Trend**	**P-Value**	**Trend**	**P-Value**
Erie	-0.71*	0.03	0.26	0.10	-0.46*	0.04
Superior	-1.40*	2.23e-3	0.87*	3.80e-5	-0.64*	0.02
Michigan	-0.99*	4.47e-4	0.69*	2.72e-4	-0.48*	4.02e-3
Huron	-1.05*	8.79e-5	0.55*	4.20e-4	-0.41*	1.10e-3
Ontario	-0.74*	2.03e-3	0.71*	8.80e-4	-0.68*	5.69e-3

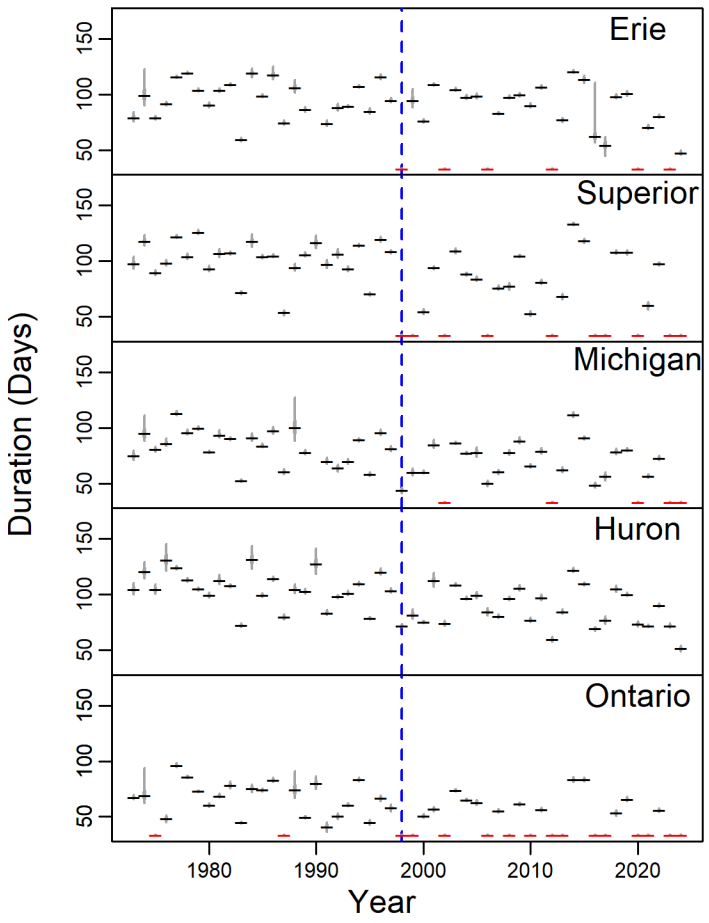


Figure 2. Duration of 10% cover each year for all five Great Lakes. The black dots represent the estimated number of days with at least 10% ice cover and the vertical grey lines represent the 95% credible interval. The red dots represent years where the model estimated coverages did not exceed 10%. The vertical blue dashed line indicates the reported regime shift year (1998). The year 1990 is not included for Lake Erie and Lake Michigan.

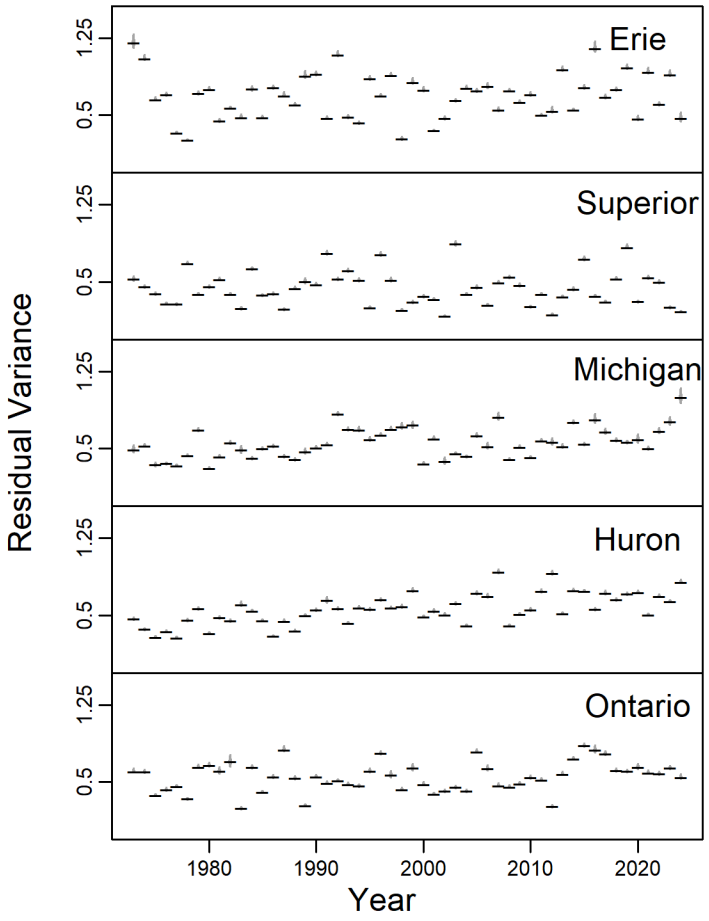


Figure 3. The residual variance for all five Great Lakes. The black dots represent the residual variance for each year and the vertical grey lines represent the 95% credible intervals.

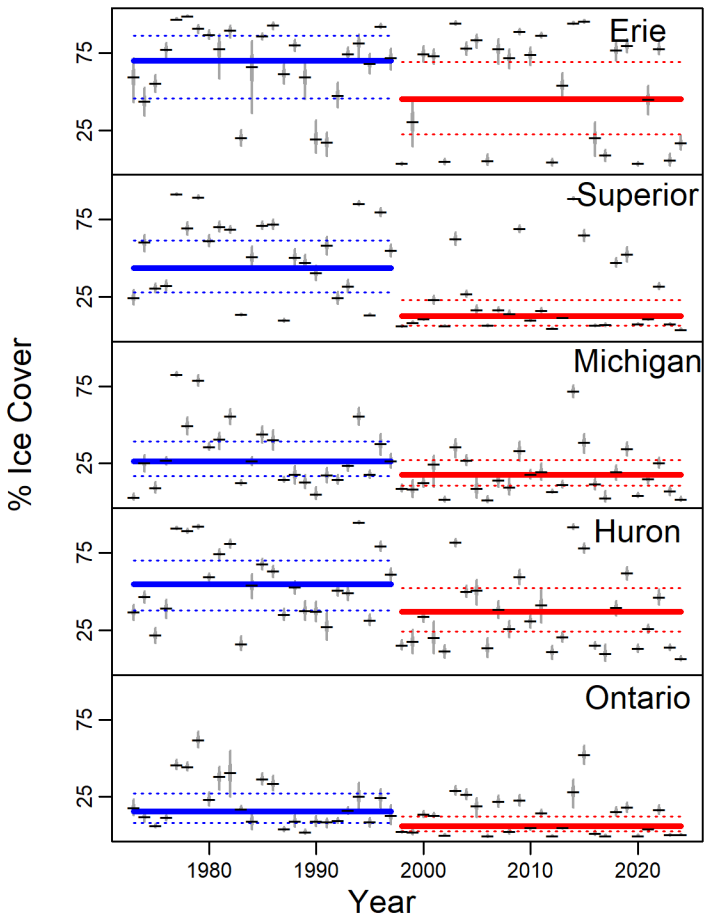


Figure 4. Peak ice cover (percent) for each year with the estimated average difference before and after 1998 for all five lakes. The blue line represents the average percent peak cover pre-regime shift (1973 – 1997) and the red line represents the average percent peak cover post-regime shift (1998 – 2024). The dashed blue and red lines represent the 95% credible intervals for the estimated average differences. The black points represent the estimated percent peak cover for each year and the vertical grey lines represent the 95% credible intervals.

4. Discussion

Unlike existing approaches that calculate individual ice cover annual metrics separately, our model captures the full seasonal pattern using a limited set of coefficients. This model allows multiple ice metrics to be derived consistently from the same underlying annual curve, allowing one to quantify how different aspects of ice cover seasonal patterns are changing over time. By pooling information across years through the BHM structure, the model also reduces uncertainty in parameter estimates.

Using this framework, we derived several key ice cover metrics and assessed long-term changes across the five Great Lakes. Although the specific rates of change for each metric vary across lakes, the general trends of change are consistent. We did not detect significant changes in the timing of peak ice cover for any lake, but all five lakes show decreases in peak coverage in recent decades. This is consistent with previous findings that ice cover across the Great Lakes is declining in both extent and duration (Assel et al., 2003; Wang et al., 2012; Mason et al., 2016). In our study, Lake Superior exhibits the largest rate of decline in peak ice cover at -0.77% per year, mostly occurred in 1998-1999 from the average of 43.61% ice cover before the El Niño to 12.74 % after. This decline may be associated with regional warming trends and Lake Superior's large size, which enhances heat retention and delays ice formation under warming conditions (Austin & Colman, 2007). Wang et al. (2012) also identified a strong decline in Lake Superior ice conditions, reporting a relative trend in annual ice coverage of −2.070% per year from 1973 to 2010 and a total reduction of 79% over the time period.

The first day reaching 10% coverage now occurs significantly later for all lakes except Lake Erie, while the last day above 10% ice cover occurs earlier for every lake, but with a higher level of uncertainty compared to the estimated first days. These results align with previous evidence that freezing dates are occurring later and ice-off dates earlier in Northern hemisphere lakes (Jensen et al., 2007; Sharma et al., 2021). Consequently, the duration of at least 10% ice cover has decreased by roughly one day per year across the basin, with Lake Superior again showing the most rapid decline (-1.40 days per year), followed by Lakes Huron, Michigan, Ontario, and Erie (-1.05, -0.99, -0.74, -0.71 days per year, respectively). This is consistent with Mason et al. (2016), who reported the largest decline in seasonal ice-cover duration for Lake Superior (0.96 days per year) and also identified significant declines for Lakes Huron, Michigan, and Ontario (−0.67, −0.55, and −0.43 days per year, respectively) but the decreasing trend for Lake Erie was not significant (−0.59 days per year). In contrast, our analysis detected a significant decreasing trend in ice cover duration for Lake Erie as well (−0.71 days per year). We detected increasing variance in ice cover for Lakes Michigan, Huron, Erie, and Ontario, suggesting that day-to-day ice conditions have become more unpredictable. Moreover, increasing variance has been identified as a potential regime shift indicator (Carpenter & Brock, 2006). Our results show decreases in ice extent and duration for the Great Lakes, and the modeling approach provides a framework to capture and compare multiple aspects of ice phenology. Overall, this model captures the general pattern of ice cover and may also be useful for assessing changing phenology in other lake processes (Dugan et al., 2024).

The advantages of using a parametric model for summarizing annual ice cover include using a consistent modeling approach across years that is reproducible, having interpretable model parameters that describe ice over patterns, and that it can be used to identify the nature of change, such as the regime shift being reflected in the peak ice cover. As all statistical models are simplifications of the real pattern, therefore, there are limitations to this modeling approach. For instance, our model is only able to characterize annual patterns that have a unimodal pattern. In our study, our model fits the data well for most of the years, but there are occasional anomalies such as observing multiple ice cover peaks which the model cannot capture (Figure S1). This caused the year 1990 for Lakes Erie and Michigan to be excluded from the 10% ice cover analyses because the first day of 10% ice cover could not be estimated. For now, the unimodal pattern is a reasonable assumption given how ice cover behaves generally in the Great Lakes region, but for patterns where unimodal is not the standard behavior, then a different parametric model would need to be proposed. Another limitation is that we removed ice cover observations of 0% from the data prior to fitting the model because of the logit-transformation. The general pattern from year to year was still captured, but a future iteration of the model may want to consider including the zeros as censored data.

Future work could include using the current BHM framework to provide short-term forecasts of Great Lakes ice cover. Using the current model as the starting point, we can incorporate the existing estimate of the average model parameters over the last 50 years as an informative guess for the upcoming year's ice cover pattern. This

initial guess can be updated when early ice season data are available, and the updated model can be used to provide a short-term forecast of ice cover until the next observations are available, and the model is updated again. After each update, we can calculate important statistics such as the first day of at least 10% cover and the peak coverage and date. This sequential updating method was used in Qian et al. (2025) to provide short-term forecasts of Lake Erie harmful algal bloom toxicity and has the potential to work well with Great Lakes ice cover too.

Given the overall success of the model for summarizing annual ice cover, our model can be used for other seasonal or phenological responses in a number of systems given that the annual pattern follows a similar structure to ice cover. This model would work for patterns that are unimodal within the defined “year.” How a “year” is defined will be system-specific. For instance, for ice cover, we used ice year (November to June) as this captures the annual ice pattern. However, one may use the U.S Geological Survey definition of water year (October to September) for hydrological data. Additionally, the BHM framework we used is an important feature of the model because it allows us to combine data from past years to evaluate changes in the seasonal pattern (Gelman et al., 2014). Ultimately, our model provides a consistent approach to estimate a smoothed peak when it might not be captured in the data very well, and it's guided by the hierarchical structure. Together, the smoothed hockey stick model and BHM approach could be useful for evaluating changes in seasonal or phenological patterns of a variety of systems.

Conflicts of Interest

The authors declare there are no conflicts of interest for this manuscript.

Data Availability

Data and computer code (in R) is publicly available at https://github.com/songsqian/Ice

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